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ORIGINAL CONTRIBUTION

Frame Shuffling Forgery Detection Method for MPEG-Coded Video

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Abstract Frame duplication and frame shuffling represent prevalent methods employed by forgers to manipulate digital video content, aiming to conceal or emphasize specific activities within the video. While several forensic techniques have been developed to address frame duplication forgery, frame shuffling forgery has received limited attention, resulting in a gap in detection methods. This paper introduces a novel forensic framework integrating motion vector analysis and scale-invariant transform features to effectively detect and locate frame shuffling forgeries in moving picture expert group (MPEG)-coded videos. The proposed algorithm's performance is rigorously assessed using precision rate, recall rate, and detection accuracy metrics across 15 test videos. Remarkably, the algorithm achieves an average detection accuracy of 100%, signifying its exceptional precision in identifying forgeries. Additionally, the algorithm demonstrates efficiency, with an average simulation time of only 36 s, underscoring its effectiveness in real-time forensic applications.

Keywords Frame duplication · Frame shuffling · Motion vector · SIFT features · Suspicious frames · MPEG-coded video

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Introduction

Since December 2019, the world has witnessed a significant shift in the way society operates, primarily due to the COVID-19 pandemic. The imposition of lockdowns led to an extensive dependence on online platforms such as Facebook, Twitter, and YouTube for information sharing and retrieving. Concurrently, the government encouraged the integration of online tools in public sectors like banking, revenue offices, and education, resulting in a continuous influx of digital content. This surge highlighted the profound impact of online sites and digital media on our society.

However, this rapid digitalization has given rise to pressing concerns regarding the security and authenticity of digital information. Malicious individuals may exploit online platforms to manipulate videos, altering crucial information, and creating confusion. Law enforcement agencies, for instance, frequently rely on CCTV footage as evidence in criminal investigations, making it essential to address the potential manipulation of these videos. Consequently, digital media, whether images or videos, cannot be considered authentic proof unless its integrity is verified by forensic experts. Hence, ensuring the authenticity of digital information has become a critical necessity. [1, 2]

The accessibility of video editing tools like photoshop and paint shop pro has further complicated matters, enabling even non-experts to manipulate videos effortlessly. In contexts such as court trials, law enforcement, defamation cases, politics, and defense planning, the authenticity of presented videos is paramount and requires meticulous examination.

To address these challenges, the field of digital forensic science plays a pivotal role. While substantial progress has been made in detecting image forgery, video forgery detection, especially in cases of interframe forgery like frame shuffling and frame duplication, presents intricate

challenges. Skilled forgers adeptly manipulate videos, leaving behind minimal forensic traces, making it challenging for experts to identify tampering. Moreover, distinguishing between compression artifacts and tampering-related inconsistencies introduced by video compression algorithms adds complexity to the detection process.

The creation of high-quality forgeries using advanced video editing tools further exacerbates the situation, blurring the line between manipulated and authentic frames. The advent of real-time video manipulation tools further compounds the challenge, especially in live or real-time video streams where reference videos are scarce.

Detecting frame shuffling and frame duplication forgeries in digital videos is challenging due to the sophisticated techniques employed by forgers. Minimal forensic traces, difficulty in distinguishing between compression artifacts and tampering-related inconsistencies, seamless blending of manipulated frames with authentic ones, and real-time video manipulation tools pose significant hurdles. Moreover, analyzing vast volumes of video data for manipulation demands substantial computational resources and time, especially in real-time applications. Additionally, forgers may employ adversarial techniques designed to deceive detection algorithms, making automated systems susceptible to identifying forgeries. [3, 4]

Addressing these challenges necessitates continuous research and the development of advanced forensic techniques. Collaboration between experts in computer vision, signal processing, and video production and editing is crucial. A multidisciplinary approach is essential to navigate the intricate landscape of video forgery detection, ensuring the authenticity and reliability of digital media in an increasingly interconnected world.

The paper is organized into five sections to provide a comprehensive understanding of the topic. In “Interframe Forgery” section, an overview of interframe forgery and its various types is presented. In “Related Work” section offers insights into existing methodologies designed to tackle frame duplication and frame shuffling forgery. The proposed framework and the dataset employed to address these issues are detailed in “Methodology” section. In “Result Analysis and Discussion” section delves into the analysis of results and performance parameters recorded during simulations. Lastly, the paper concludes by evaluating the efficiency and accuracy of the proposed method in “Conclusion” section.

Interframe Forgery

Interframe forgery, often referred to as temporal forgery, encompasses the manipulation of frames within a video, disrupting its chronological sequence and distorting its time-domain representation. This manipulation gives rise to a variety of techniques falling under the umbrella term of interframe forgery. Figure 1 provides a visual representation of the different categories of interframe forgery [3, 4].

a. *Frame Deletion Forgery*
This technique involves the deliberate removal of specific frames from the video, with the intention of skipping particular activities captured in the footage.

b. *Frame Insertion Forgery*
Frames sourced from the same video or different origins are inserted at randomly chosen locations within the target video. This manipulation serves to integrate specific scenes into the video, thereby altering its context and narrative.

c. *Frame Duplication Forgery*
Frames from one specific time frame within the video are duplicated and inserted in the same sequential order at another time frame. This method is employed to either conceal or emphasize specific activities within the video content.

d. *Frame Shuffling Forgery*
In this case, frames are rearranged and then pasted into different time frames. Shuffling disrupts the natural order of frames, leading to a distorted storyline within the video. Shuffling can be executed in two ways:

- Copying and Pasting Frames in Reverse Order
- Randomly Copying and Pasting Frames at Different Locations within the Same Video

Detecting such forgeries poses a significant challenge due to the fact that the manipulated frames originate from the same video, maintaining consistent statistical properties. As a result, distinguishing these forged frames from authentic content becomes a complex task.

This study focuses specifically on frame shuffling forgery within MPEG-4 digital videos. The effectiveness of proposed detection method is assessed through experimentation conducted on manipulated videos employing any of the techniques mentioned above [4, 5].

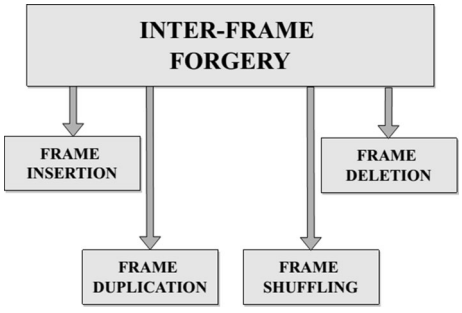


Fig. 1 Classification of interframe forgery

Related Work

Substantial research has been done to address the interframe forgery detection problem. Most of the methods reported in the literature has addressed frame deletion and insertion forgery detection techniques. Relatively little work is focused on frame duplication and frame shuffling forgery detection problem. This section provides an overview of recent methodologies developed to tackle the challenges posed by frame duplication and frame shuffling forgery. The review is provided in tabular form, which covers the parameters like method of detection and detection accuracy (Table 1).

a. ***Research Gaps***

a *High Computational Time*: Many existing methods suffer from excessively long computational times, hindering their practical applicability.

b *Low Detection Accuracy for Moving Camera Scenes*: Certain techniques struggle to maintain high detection accuracy, particularly in scenarios involving moving cameras, limiting their effectiveness.

c *Lack of Marking Duplicate Frame Locations*: A significant drawback of several methods is their inability to accurately mark the locations of duplicate frames within the manipulated video content.

d *Limited Detection Methods for Irrespective Order and Total Count*: Only a handful of methods in the literature can detect duplicate frames regardless of their order and the total count of forged frames.

In light of these challenges, this paper introduces a novel and robust passive blind algorithm designed to detect frame shuffling forgery, irrespective of the order and number of forged frames. This proposed method aims to overcome the limitations present in the existing techniques, providing a promising avenue for enhancing the accuracy and efficiency of interframe forgery detection in the field of video forensics.

Methodology

The proposed approach consists of two distinct steps. Firstly, it detects suspicious frames by analyzing the motion vectors of the frames. Secondly, a validation process is employed to confirm these suspicious frames as forged frames, utilizing SIFT features.

Table 1 Survey of existing methods

Sr. No	Author	Method of detection	Attack addressed	Detection accuracy
1	Fadl et al. [5]	Motion energy image HOG	Frame duplication Frame shuffling	96%
2	Aishwarya. et al. [6]	Optical flow coefficient	Frame duplication Frame shuffling	94%
3	Ulutas et al. [7]	Binary features	Frame duplication Frame mirror	99.35%
4	Ulutas et al. [8]	Bang of words	Frame duplication	97.54%
5	G. Ulutas et al. [9]	Correlation coefficient	Frame duplication	91%
6	Bozkurt et al. [10]	DCT coefficients	Frame duplication	98.64%
7	Fadl et al. [11]	Temporal average gray level co-occurrence matrix	Frame duplication Frame shuffling	96%
8	Ulatus et al. [12]	Local difference binary PSNR	Frame duplication	96.43%
9	Mohiuddin et al. [13]	Structural Similarity Index	Frame duplication	99%
10	Ren et al. [14]	Levenshtein distance	Frame duplication	99.5%
11	Chengjiang Long et al. [15]	Machine Learning	Frame duplication	85%
12	Liba Manopriya et. al. [16]	Discrete wavelet transform HOG features	Frame duplication, Frame deletion	84% 91%
13	Fadl et al. [17]	Gaussian RBF multi-class support vector machine (RBFMSVM)	Frame duplication	98.5%
14	Oraibi et al. [18]	3D-CNN	Frame duplication Frame shuffling	99%
15	Maryam Munawar et.al [19]	deep learning framework consisting of Inflated 3D (I3D) and Siamese-based Recurrent Neural Network (RNN)	Frame duplication	93%
16	P. Priyanka et. al. [20]	Matrix decomposition using RPCA	Frame Duplication	–

Step 1: Detection of Suspicious Frames

Suspicious frames are the frames that are potential candidates for shuffling forgery. For instance, consider a scenario where a set of five frames from time frame Tn is copied, shuffled, and inserted at another location in time frame Tn+2. This manipulation disrupts the natural sequence of frames. The discontinuity among frames is scrutinized by calculating the mean values of frames. An abrupt variation in the mean values of duplicate frames, compared to authentic frames, is observed. The proposed method utilizes this sudden variation in mean values to identify the suspicious frames, which are likely subjected to shuffling forgery.

Steps to detect suspicious frames:

1. *Convert test video into frames*
   - Let $f(n)$ is the sequence of the frames of test video indicated by Eq. (1).

$$f(n) = \{f1, f2, f3, f4, f5, \ldots, fn\} \tag{1}$$

2. Find motion vector of each frame
   - The motion vector of each frame is calculated using three step block matching algorithm represented by Eq. (2).

$$x(n) = \{x1, x2, x3, x4, x5, \ldots, xn\} \tag{2}$$

   where $x1, x2, x3, x4, x5, \ldots \ldots .., xn$ represents motion vector images.
3. Select I-Frame as a marker frame. (Here, $f1$ is considered as I-Frame.)
4. Next subtract each motion vector image from I-frame as indicated by the Eq. (3).

$$y(n) = \{f1 - x(n-1)\} \tag{3}$$

   hence,

$$y(n) = (y1, y2, y3, \ldots, yn) \tag{4}$$

   where $y1$, $y2$, $y3$ are the difference images indicated by Eq. (4).
5. Find the mean of all the difference frames shown by Eqs. (5 and 6).

$$M(n) = \text{mean}(y(n)) \tag{5}$$

$$M(n) = (m1, m2, m3, \ldots, mn) \tag{6}$$

   where $m1$, $m2$, $m3\ldots$ represent the mean values of difference frames.
6. Compare the mean value of all the frames stored in $M(n)$.
7. Finally, the frames with same mean values are identified and declared as suspicious frames.

Assume that, $m5$ and $m10$ are marked as suspicious frames i.e., but

$$m5 = m10$$

$$m5 = y5 \text{where} y5 = (f1 - x6) \tag{7}$$

$$m10 = y10 \text{where} y10 = (f1 - x11) \tag{8}$$

8. From Eq. 7 and 8, we get three frames, namely $f1, \times 6, \times 11$. As $f1$ is marker frame, it will be ignored and remaining frames $\times 6$ and $\times 11$ are marked as suspicious and stored in vector called as ssp.

Step 2: Validation Step

Validation is the second crucial stage in the process. During this step, the previously identified suspicious frames undergo validation using SIFT features. Following steps are employed for validation:

1. The SIFT features of all frames are extracted and then compared using the cosine similarity index to confirm the matching between frames.
2. Subsequently, the RANSAC homography algorithm is applied to the matched features, effectively reducing the false positive rate of detection.
3. Lastly, the original frame numbers and the corresponding shuffled frame numbers are displayed, providing a clear overview of the validation results.

Step 3: Distinguish Between Authentic and Forged Frames

In the final step, the task is to differentiate between genuine and manipulated frames. To achieve this, each manipulated frame is subtracted from its preceding frame, and this variance is saved in two distinct vectors. The sorting process employs the following logic:

1. Every identified suspicious frame is subtracted from its preceding frame and the resulting differences are stored in variables p and q, as represented by Eqs. 9 and 10.

$$p(i) = (x5 - x6) \tag{9}$$

$$q(i) = (x10 - x11) \tag{10}$$

2. Ultimately, if the value of $p(i)$ is greater than $q(i), \times 6$ is identified as the manipulated frame, and vice versa.

5. Result Analysis and Discussion

The performance of the proposed forensic scheme is tested in two phases. In first stage, detection accuracy of proposed method is calculated while second stage, the total simulation time required to locate the forged frames is recorded. This section is further divided into three subsections, which are discussed one by one to demonstrate the effectiveness and detection ability of proposed method.

Details of Test Video

The proposed method was tested using a set of 15 MPEG-coded test videos, all encoded at a frame rate of 30 FPS and featuring various resolutions, ranging from 320×240 to 1280×720 pixels. Among these videos, ten were sourced from websites URL mentioned in references [21] and [22], while the remaining five were obtained from YouTube and manipulated to simulate the second case scenario. Within this set, eight videos were captured using a moving camera, while seven were recorded using a stationary camera. Additional information about each video can be found in the Table 2 provided below.

Performance Parameters

To assess the algorithm's performance, three key metrics were employed as given below.

Precision Rate: This metric measures the likelihood of correctly identifying a forgery.

$$PR = \frac{TP}{TP + FP} \tag{11}$$

Recall Rate: This metric indicates the probability of detecting forged frames accurately.

$$RR = \frac{TP}{TP + FN} \tag{12}$$

Detection Accuracy: Represents the accurately detected forged frames.

$$DA = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{13}$$

True Positive(TP) :Authentic is detected as authentic

True Negative(TN) : Forged is detected as forged

False Positive(FP) : Authentic is detected as forged

False Negative(FN) : Forged is detected as authentic

Simulation Results

The proposed method begins by identifying frames that appear suspicious and may be indicative of frame shuffling forgery. Subsequently, these identified suspicious frames undergo validation to confirm their status as forged frames, utilizing SIFT features and RANSAC homography.

Table 2 Details of dataset

Sr. No	Test video	No of frames before attack	No of frames after attack	Total no of forged frames	Nature of attack
1	T1	319	381	62	Frame no 157–220 pasted at 50–112
2	T2	329	373	44	Frame no 104–60 pasted at 70–214
3	T3	314	452	138	Frame no 258–120 pasted at 314–452
4	T4	321	376	55	Frame no 56–1 pasted at 57–113
5	T5	34	34	09	Frame no 3–7 pasted at 25–21 Frame no 15–16 pasted at 29–28
6	T6	330	547	217	Frame no 547–330 pasted at 1–218
7	T7	425	616	191	Frame no 351–160 pasted at 425–616
8	T8	430	530	100	Frame no 280–180 pasted at 6–106
9	T9	430	530	100	Frame no 200–100 pasted at 300–400
10	T10	250	300	50	Frame no 110–10 pasted at 150–250
11	T11	400	405	05	Frame no 70–75 pasted at 300, 255, 450–453
12	T12	400	450	30	Frame no 130–100 pasted at 400–410, 200–210, 310–300
13	T13	412	452	40	Frame no 90–50 pasted at 186–210, 230–204
14	T14	274	293	19	Frame no 50–70 pasted at 110–100, 90–95, 240, 230, 160
15	T15	50	55	08	Frame no 5–10 pasted at frame no 25, 49, 21, 30, 35 Frame no 11–15 pasted at 54–50

One of the key strategies employed is the assessment of motion vectors within consecutive frames. In general, the analysis of motion vectors for consecutive frames reveals a distinct pattern of consistency. Notably, genuine video frames exhibit minor variations in their motion vectors. However, when frames within the video sequence are subjected to shuffling, the motion vectors within that segment display significant deviations. This study aims to capture these motion vector variations to identify the shuffled frames. This is accomplished by calculating the mean value of motion vectors across frames and used it to detect suspicious frames. This process is exemplified in Fig. 2, where the motion vectors of five consecutive frames from test video T5 are depicted. These images clearly demonstrate that there is no significant change in motion vectors, indicating the authenticity of these frames.

Further insights into the validation process are provided in Fig. 3, which presents SIFT keypoints for selected frames from test videos T1, T5, T15, T13, and T14. In the case of test video T1, frames 50–112 have been copied and pasted to the location of frames 220–157. This operation is visually represented in Fig. 3a, where the recurring pattern in the bar graph, denoted by a black circle, confirms the precise correspondence between the SIFT keypoints of original and forged frames. In the context of test video T5, frames 3–7 are pasted at the location of frames 25–21, and frames 15–17 are pasted at 29–27, demonstrating the occurrence of multiple forgeries within a single video. The former scenario reflects a frame duplication forgery, while the latter illustrates a frame shuffling forgery. The consistent repetition of this mirrored pattern can be observed in Fig. 3b.

As for test video T15, the frames are deliberately shuffled to different locations. Figure 3c showcases similar keypoints for both the original and the forged frames, highlighted by blue circles, confirming that the proposed method successfully identifies these frames, despite their altered order and placement.

Additionally, Fig. 3d represents SIFT keypoints for a brightness variation attack, wherein original frames located at 400–460 are duplicated, their brightness values manipulated, and then these frames are shuffled and pasted at the location 70–130. Subsequent experimentation involves a combination of blur and brightness variation attacks applied to the same frame. The results are presented in Fig. 3e, where the original and the attacked frames are situated at locations 351–160 and 425–616, respectively.

Furthermore, the study explored the resilience of algorithm against diverse attacks, including brightness and blur variations. Figure 3d showcased the successful identification of manipulated frames subjected to brightness variation attacks, wherein frames located at 400–460 were copied, brightness values altered, and the sequence reshuffled at the location 70–130. Additionally, the combination of blur and brightness variations in the same frame was meticulously examined. The results, depicted in Fig. 3e, where the original and the attacked frames are situated at locations 351–160 and 425–616, respectively, revealed the algorithm's ability to discern manipulated frames even under complex, multifaceted attacks, reaffirming its robustness and effectiveness.

This analysis concludes in the observation that the SIFT keypoints of original and duplicate frames exhibit an exact match, confirming the presence of frame shuffling forgery and specifying its location. Furthermore, it is evident that the developed algorithm proves robust against a variety of attacks, including brightness variation and blur variation, camera position variation, as these manipulations do not affect the stability of SIFT keypoints and, consequently, the detection accuracy.

In Fig. 4, provides a visual representation of the matching between authentic and forged frames, with the original frame displayed on the left and the shuffled frame on the right. The

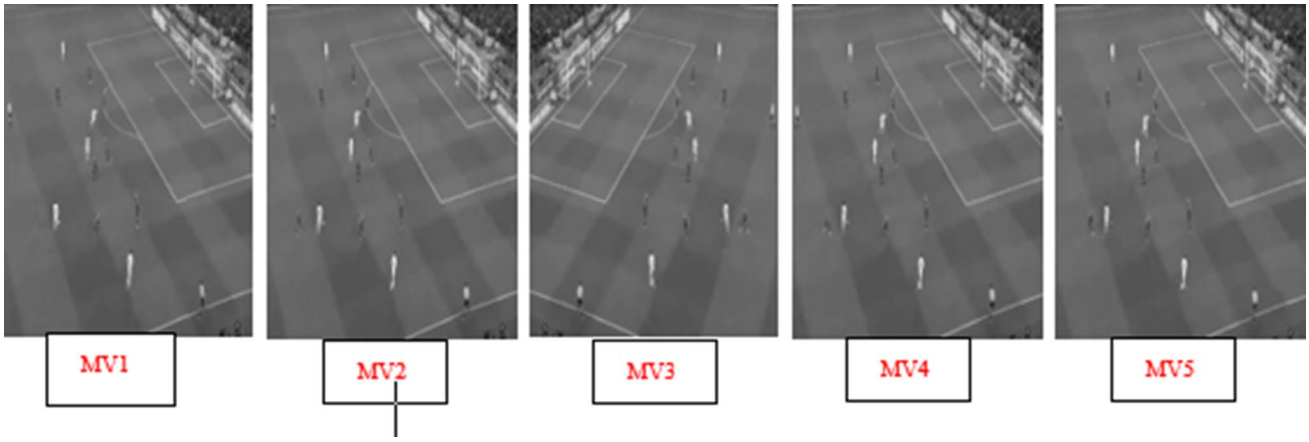
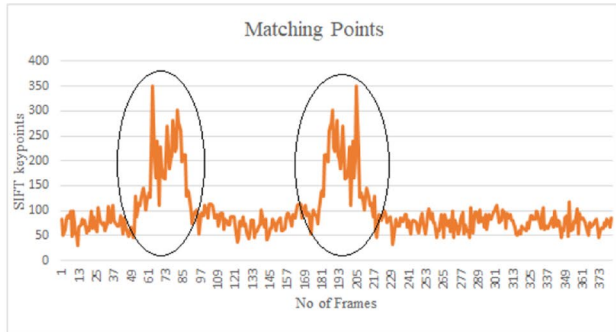
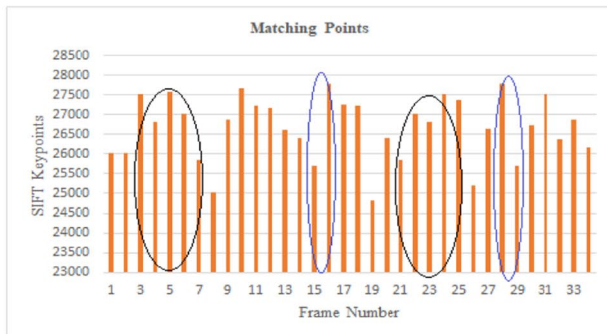


Fig. 2 Motion vector of successive frames of video T1

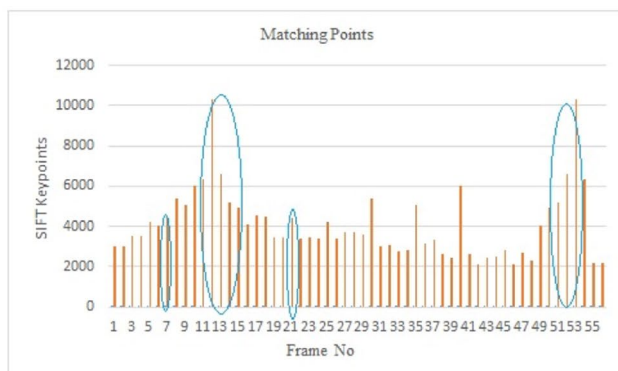
Fig. 3 **a** SIFT keypoints for frame shuffling forgery for T1. **b** SIFT keypoints for multiple frame shuffling forgery for T5. **c** SIFT keypoints for frame shuffling forgery for T15. **d** SIFT keypoints for frame shuffling forgery with brightness variation attack for T12. **e** SIFT keypoints for frame shuffling forgery with blur and brightness variation attack for T11



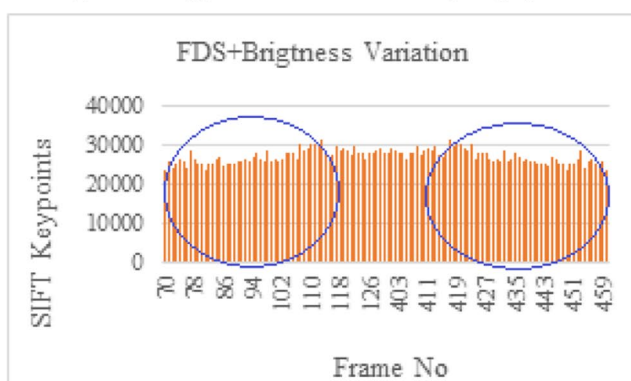
(a). SIFT keypoints for frame shuffling forgery for T1



(b). SIFT keypoints for multiple frame shuffling forgery for T5

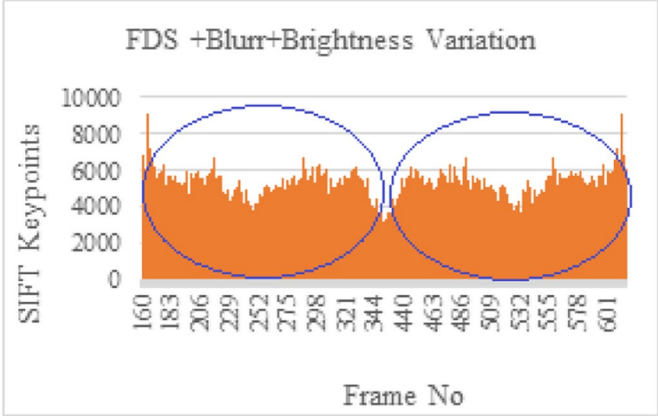


(c). SIFT keypoints for frame shuffling forgery for T15



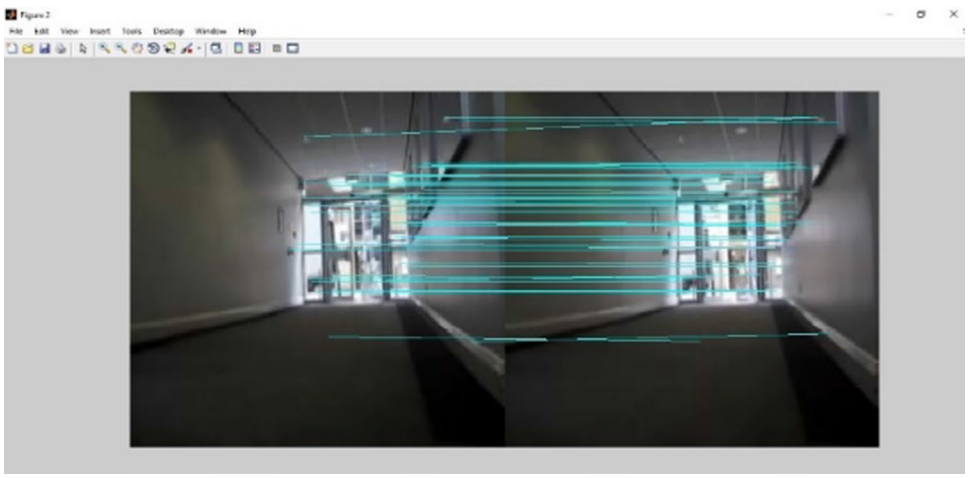
(d). SIFT keypoints for frame shuffling forgery with brightness variation attack for T12

Fig. 3 (continued)



(e). SIFT keypoints for frame shuffling forgery with blurr and brightness variation attack for T11

Fig. 4 Matching between authentic and forged frame



alignment between the two is illustrated by the presence of blue lines connecting corresponding keypoints, underscoring the successful detection of video forgeries through proposed method.

Performance Analysis

To assess the efficacy of the proposed system, a comprehensive set of experiments was conducted on a total of 15 manipulated videos, involving the analysis of around 5000 frames to detect instances of forgery. This section provides an in-depth performance analysis of the proposed system, concentrating on three pivotal parameters: precision rate (PR), recall rate (RR), and detection accuracy (DA).

In the preliminary stage of result analysis, essential metrics including true negatives (TN), true positives (TP), false negatives (FN), and false positives (FP) were meticulously calculated. These metrics served as the basis for computing PR, RR, and DA. The detailed results of these computations are presented in Table 3, where PR, RR, and DA values for all test videos are showcased. Remarkably, the values displayed in the table consistently attain 100% across all performance parameters. This remarkable uniformity underlines the robustness and trustworthiness of the proposed method, demonstrating an average detection accuracy of 100%. Additionally, the average simulation time required for identifying forged frames was determined to be 36 s.

Crucially, the consistent 100% performance observed across all test videos highlights the proposed method's resilience in the face of varying camera positions and different attacks, including brightness and blur variations. Regardless of the camera's orientation or the specific image processing operations applied to the frames, the system's performance remains consistently high. This outcome underscores the method's ability to sustain its effectiveness and precision, irrespective of the camera's position or the nature of attacks, reaffirming its robustness in real-world applications.

Table 3 Performance parameters

Sr. No	Video	PR (%)	RR (%)	DA (%)	Avg PR	Avg RR	Avg DA	Avg simulation time
1	T1	100	100	100	100%	100%	100%	36 s
2	T2	100	100	100				
3	T3	100	100	100				
4	T4	100	100	100				
5	T5	100	100	100				
6	T6	100	100	100				
7	T7	100	100	100				
8	T8	100	100	100				
9	T9	100	100	100				
10	T10	100	100	100				
11	T11	100	100	100				
12	T12	100	100	100				
13	T13	100	100	100				
14	T14	100	100	100				
15	T15	100	100	100				

Table 4 Comparison with existing methods

Sr. No	Name of author	Attacks addressed	Camera position	Precision rate	Recall rate	Detection accuracy
1	Sondos Fadl et. al. [5]	Brightness and blur variation	Stationary	96	94	94
2	Aishwarya. et al. [6]	No attacks considered	Stationary	95	93	96
3	Guzin Ulutas et. al. Beste Ustubioglu et.al [9]	No attacks considered	Stationary & moving	99.50	99.30	99.10
4	Sondos Fadl et. al. [11]	Brightness and blur variation	Stationary	94	98	99
5	Proposed method	Brightness and blur variation	Stationary & moving	100	100	100

Comparison with Existing Methods

To assess the effectiveness of developed system, a comprehensive comparison is conducted with similar methodologies reported in the literature. This comparative analysis encompassed essential aspects such as detection capability metrics, the types of image processing operations applied to shuffled frames, and the camera positions of the test videos used for experimentation.

In our evaluation, we reviewed the latest four relevant studies from the literature. The findings of this comparative analysis are summarized in Table 4, which provides the percentage values for precision rate, recall rate, and detection accuracy for each method under consideration. A detailed examination of the data in the table clearly demonstrates that developed system consistently outperforms other methods cited in the literature. Specifically, developed system exhibits the highest detection accuracy, precision rate, and recall rate when compared to its counterparts, highlighting its superior performance.

Furthermore, Fig. 5 visually represents this comparative analysis between our developed system and existing methodologies. The graphical depiction confirms the results presented in Table 4, providing a visual confirmation of developed system's superior performance in terms of detection accuracy, precision rate, and recall rate.

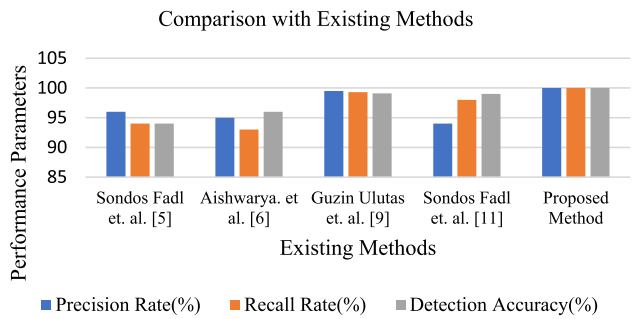


Fig. 5 Comparison with existing methods

This thorough comparison highlights the robustness and reliability of our developed system, establishing it as a robust solution in the field of video forgery detection. The findings not only highlight its superiority over existing

methods but it as an innovative and effective approach in the field of video forgery detection techniques.

Conclusion

In summary, this research paper introduces a passive blind algorithm designed specifically for detecting frame shuffling forgery in MPEG-4 digital videos. The approach involves a sophisticated framework integrating motion vectors (MV) and scale-invariant feature transform (SIFT) features. The experimental phase included a diverse range of 15 test videos, each varying in configuration parameters such as resolution, frame rate, and frame length. To enhance the complexity of the detection process shuffled frames are manipulated using various image processing operations, including brightness and blur variations, before integrating them into different time frames of the video. Additionally, multiple attacks are introduced within a single video, such as frame shuffling, frame duplication together in one video, further complicating the detection process.

To assess the efficacy of proposed method, three crucial performance metrics are used: precision rate (PR), recall rate (RR), and detection accuracy (DA), comparing the results with existing techniques in the literature. The outcomes of our study were exceptionally promising, demonstrating an overall detection accuracy of 100%. This remarkable performance not only highlights the efficiency of proposed system but also underscores its resilience in identifying frame shuffling forgery amidst various complex attacks in digital videos.

This research signifies a significant advancement in the field of video forgery detection, contributing substantially to the development of cutting-edge video forensics techniques.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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